

1. Introduction

Drug use and smoking among young people pose serious social and public health issues. These actions can create patterns that continue into adulthood and have long-term negative effects on one's health. To develop interventions aimed at reducing the prevalence of these habits, especially among vulnerable youth populations, it is imperative to comprehend the factors that contribute to them.

The main question our study aims to address is: **What are some of the major factors contributing to smoking and drug use in youth?** Finding connections between important factors like age, socioeconomic status, family background, parental supervision, mental health, and peer influence, as well as identifying at-risk groups, are urgent needs that are addressed by this question. Our objective is to offer evidence-based insights to direct prevention strategies by examining these variables.

This project uses rigorous statistical analyses to investigate these relationships using a Kaggle dataset that gathers data on youth drug use and smoking from 2020 to 2024. Numerous variables that capture the demographic, socioeconomic, and psychological aspects of substance use are included in the dataset. By conducting these analyses, we hope to find significant trends and connections that will eventually advance our knowledge of this crucial problem.

2. Data

Sources: Our data summarizes the statistics of youth smoking and drug use for the years between 2020 to 2024, age groups from 10 to 80+ grouped by decades except 10-19 years olds are separated into two groups of 5 years. There also includes the socioeconomic status of the different age groups for different years, influence of family or other peers, as well as if family background or if they have any mental illness that might affect the individual and other variables indicated in the data. The summary data was compiled by Waqar Ali, from Kaggle, who got all the data from reputable sources and research studies, surveys and other reports. This dataset only contains information from years 2020-2024.

Variables: The data includes information for each age group for the years of 2020 to 2024, as well as smoking prevalence for each age group, peer influence, family background/parental supervision, their socioeconomic status, mental health and drug experimentation.

Summary: The Youth Smoking and Drug Dataset includes 10,000 observations of 15 variables described in the section above. The dataset includes 6 categorical variables and 9 numeric variables. The variables that are used in this report are summarized in Table 1. This includes basic summary statistics including mean, standard deviation, and frequency of the variables used. The average smoking prevalence of the groups in this dataset is

27.44% and the standard deviation is 12.98%. The average drug experimentation is 40.15% and the standard deviation is 17.52%.

		Age_Group										All
		10-14	15-19	20-24	25-29	30-39	40-49	50-59	60-69	70-79	80+	
Smoking_Prevalence	Mean	28.09	27.44	27.43	27.27	27.67	27.29	27.52	27.01	27.25	27.40	27.44
	Std	12.83	13.06	12.97	12.57	13.14	13.34	12.77	13.14	13.09	12.83	12.98
Peer_Influence	Mean	5.28	5.54	5.52	5.32	5.27	5.55	5.63	5.38	5.54	5.37	5.44
	Std	2.84	2.84	2.83	2.88	2.93	2.83	2.89	2.87	2.87	2.82	2.86
Family_Background	Mean	5.65	5.54	5.47	5.44	5.57	5.44	5.44	5.59	5.52	5.48	5.51
	Std	2.89	2.79	2.93	2.91	2.85	2.87	2.87	2.84	2.84	2.86	2.87
Parental_Supervision	Mean	5.43	5.44	5.60	5.47	5.49	5.62	5.40	5.69	5.59	5.56	5.53
	Std	2.86	2.93	2.92	2.87	2.92	2.92	2.92	2.83	2.90	2.84	2.89
Mental_Health	Mean	5.49	5.52	5.42	5.55	5.32	5.54	5.40	5.44	5.60	5.42	5.47
	Std	2.86	2.91	2.86	2.78	2.91	2.87	2.89	2.87	2.88	2.95	2.88
Drug_Experimentation	Mean	40.64	40.30	39.87	40.66	40.45	40.14	39.56	40.10	38.83	40.90	40.15
	Std	17.17	17.64	17.44	17.79	17.73	17.63	17.41	17.47	17.43	17.42	17.52
Socioeconomic_Status												
High	N	309	366	345	336	348	332	336	338	310	344	3364
	ColPctN	30.69	36.09	33.01	34.50	34.08	33.17	33.57	33.14	33.05	35.14	33.64
Low	N	347	347	321	303	333	323	318	359	314	327	3292
	ColPctN	34.46	34.22	30.72	31.11	32.62	32.27	31.77	35.20	33.48	33.40	32.92
Middle	N	351	301	379	335	340	346	347	323	314	308	3344
	ColPctN	34.86	29.68	36.27	34.39	33.30	34.57	34.67	31.67	33.48	31.46	33.44

Table 1: A summary of the variables used in this analysis of the Youth Smoking and Drug Dataset

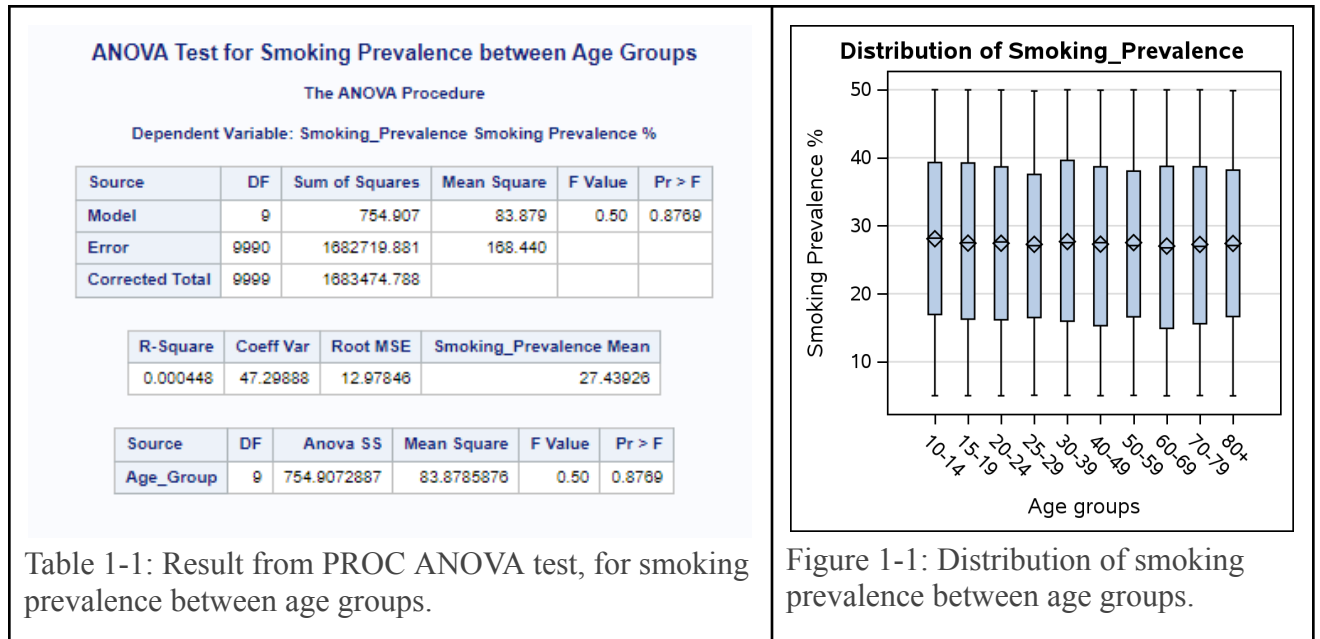
3. Specific Questions

a. Specific Question 1 (Chenchenyi Zhu)

After seeing a high number of adolescents and youths engaging in smoking and drinking, some speculate that there might be some factors that might affect a persons' behavior and ways of thinking that might cause one to engage in behaviors that are unhealthy in the long run and last for a long time. In this section, we answer the question: which age group has a higher chance in engaging in smoking?

- i. Methods: To see which age group has a higher chance in engaging in smoking, we have to compare the average means of smoking prevalence of each age group of all years by conducting an ANOVA test. Using the null hypothesis that the mean smoking prevalence of each age group is equal to each other, and the alternative hypothesis that the mean smoking prevalence between age groups is different.
- ii. Results: From the data Table 1-1 results shown below, it can be seen that the mean average smoking prevalence for all age groups is to be around 27.43926. However, from Table 1-1, the p-value of the result is 0.8769, which is greater than the alpha value (0.05). The conclusion to this ANOVA test is, we would not reject the null hypothesis that the mean smoking prevalence of each age group is equal to each other, since p-value is greater than alpha value. Since we have large n, more than 10,000 data points, it makes it easier to detect small differences in treatment means, so these results would make a good conclusion to the analyzed hypothesis. We can also see from Figure 1-1 that the smoking prevalence percentage

of the age groups are closely about the same, and there is not much difference.



b. Specific Question 2 (Abby Kovach)

In order to identify the groups that are at the most risk of engaging in smoking or drug use, one variable that may be helpful to look at is socioeconomic status. Socioeconomic status is used to identify a person's economical resources and social position. It is typically divided into three groups: low, middle, and high. By analyzing if these groups have a significant difference in smoking prevalence or drug experimentation, we may be able to identify those who are most at risk of engaging in these behaviors.

- i. Methods: In order to determine whether or not socioeconomic status is related to smoking prevalence or drug experimentation rates, we will compare the means of the rate of these behaviors by status level (low, middle, high) via one-way ANOVA. We will use a null hypothesis that the mean rate of these behaviors is equal across socioeconomic status levels. Using two PROC ANOVA statements in SAS for smoking prevalence and drug experimentation rates, we will be able to determine if there is a statistically significant difference in the means of these behaviors based on status level.
- ii. Results: According to table 2-1, ANOVA found no significant difference in the means of smoking prevalence between socioeconomic status levels. Due to the high P-Value (0.4791), we fail to reject the null hypothesis which states that the mean of smoking prevalence rates across socioeconomic status levels are equal. This is also apparent when plotting the distribution of smoking prevalence in figure 2-1 where we see very little difference in between status levels. Similarly, ANOVA found no

significant difference in the means of drug experimentation rates as seen in table 2-2. The P-Value of this test (0.7088) is too high to allow us to reject the null hypothesis that the mean drug experimentation rates across status levels are the same. Due to 10,000 observations being present in this dataset, the sample size is sufficient to analyze the results of ANOVA. The normality of these distributions may come into question, as they don't appear to be strictly normally distributed. In figures 2-3 and 2-4, the kernel density of both smoking prevalence and drug experimentation appears to be symmetrical, but perhaps too concentrated under the center of the curve, leaving the tails too small.

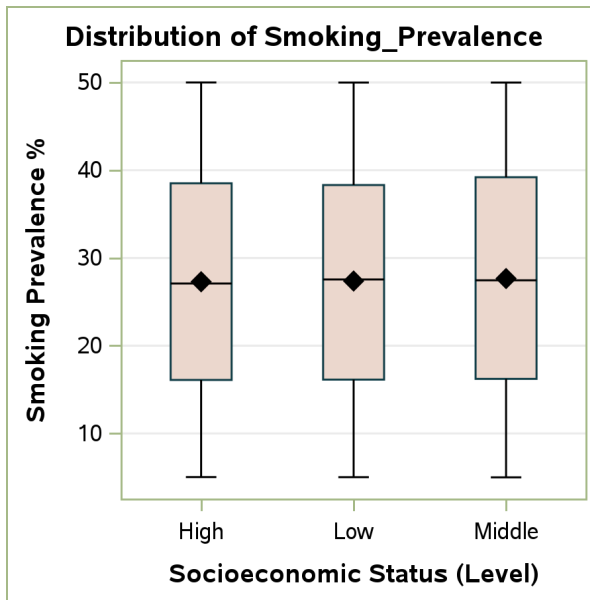


Figure 2-1: The distribution of smoking prevalence across socioeconomic status level

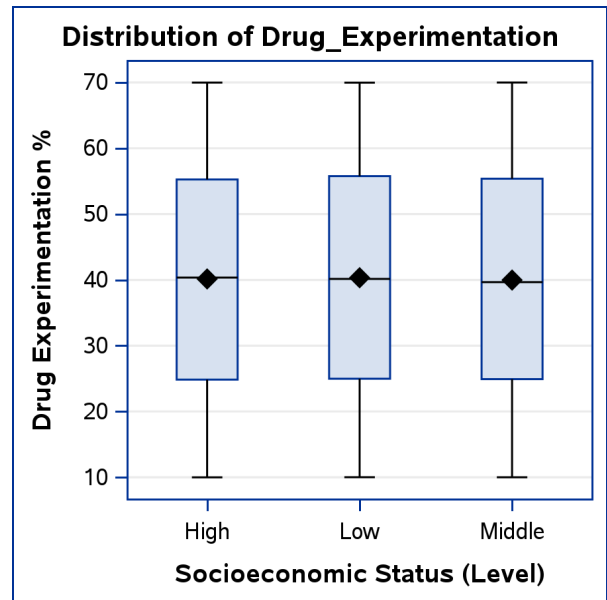


Figure 2-2: The distribution of drug experimentation across socioeconomic status level

Smoking Prevalence by Socioeconomic Status ANOVA					
The ANOVA Procedure					
Dependent Variable: Smoking_Prevalence Smoking Prevalence %					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	247.780	123.890	0.74	0.4791
Error	9997	1683227.008	168.373		
Corrected Total	9999	1683474.788			

R-Square	Coeff Var	Root MSE	Smoking_Prevalence Mean
0.000147	47.28944	12.97587	27.43926

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Socioeconomic_Status	2	247.7803547	123.8901773	0.74	0.4791

Table 2-1: Results from PROC ANOVA related to smoking prevalence across socioeconomic levels

Drug Experimentation by Socioeconomic Status ANOVA					
The ANOVA Procedure					
Dependent Variable: Drug_Experimentation Drug Experimentation %					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	211.228	105.614	0.34	0.7088
Error	9997	3067555.489	306.848		
Corrected Total	9999	3067766.717			

R-Square	Coeff Var	Root MSE	Drug_Experimentation Mean
0.000069	43.62886	17.61707	40.15018

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Socioeconomic_Status	2	211.2282948	105.6141474	0.34	0.7088

Table 2-2: Results from PROC ANOVA related to drug experimentation across socioeconomic levels

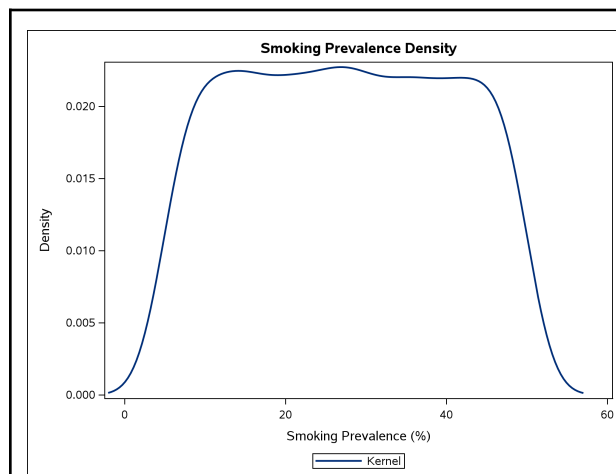


Figure 2-3: Kernel density of smoking prevalence

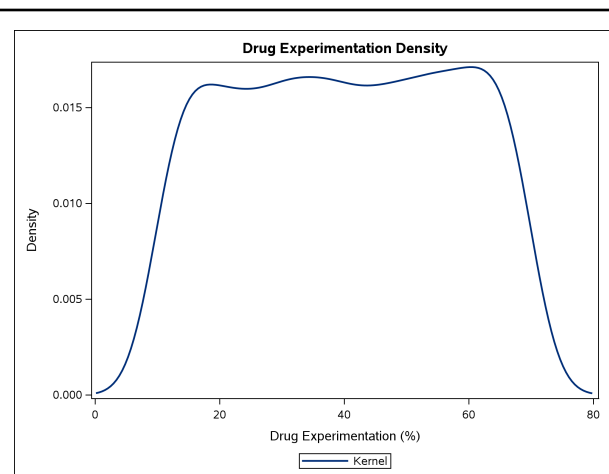


Figure 2-4: Kernel density of drug experimentation

c. Specific Question 3 (Yulai Zhang)

This study investigates the relationship between parental supervision levels and family background and the prevalence of smoking among young people. Among the variables examined are Parental_Supervision, a numerical variable that gauges parental supervision, and Family_Background, a categorical variable that represents the consistency of family support. The main objective is to ascertain whether there are notable differences in smoking prevalence between these levels, both separately and in combination. Estimating mean prevalence rates and utilizing the proper statistical techniques to check for significant differences are the goals of inference.

- Methods: ANOVA was used to compare the mean smoking prevalence across various levels of each variable in order to evaluate the effects of

parental supervision and family background. For every group, PROC TABULATE produced descriptive statistics like means and confidence intervals. The hypothesis that there are significant differences in smoking prevalence between groups was tested using PROC ANOVA.

Additionally, the interaction effects between parental supervision and family background were investigated using a generalized linear model (GLM). The results were displayed using visualizations made with PROC SGPLOT, such as bar charts and stacked plots. Among the hypotheses examined are:

Null Hypothesis: The prevalence of smoking is not substantially affected by parental supervision levels or family background.

Alternative Hypothesis: The prevalence of smoking varies greatly among these levels.

- ii. Results: There were no statistically significant differences in smoking prevalence between parental supervision levels ($F(9, 9990) = 0.75, p = 0.66$) or family background levels ($F(9, 9990) = 0.70, p = 0.71$), according to the ANOVA results. Likewise, there was no significant interaction between these variables ($F(81, 9900) = 0.80, p = 0.90$). Nonetheless, descriptive statistics showed minor differences: lower parental supervision levels tended to correlate with slightly higher prevalence rates, while Group 6 of Family_Background had the highest mean prevalence (28.22%, 95% CI: 27.42–29.02). These trends were emphasized by the stacked bar chart that showed the prevalence of smoking by parental supervision within the family background, but it did not demonstrate any significant interactive effects.

According to the results, smoking prevalence is not significantly influenced by parental supervision levels or family background, either alone or in combination. The validity of statistical tests was guaranteed as the assumptions of homoscedasticity and normality were satisfied.

Limitations include the cross-sectional nature of the data, which makes it impossible to draw conclusions about causality, and possible confounders that were not taken into consideration throughout the research. To properly understand these associations, future studies should examine longitudinal data and include more variables.

- iii. Pictures:

Smoking Prevalence by Family Background

Summary Statistics for Family Background		Mean	Std	95_LCLM	95_UCLM	All
Smoking_Prevalence	Family_Background					Sum
	1	27.28	12.85	26.49	28.07	27714.91
	2	27.00	13.12	26.16	27.84	25513.29
	3	27.57	12.80	26.78	28.37	27515.99
	4	27.57	13.41	26.74	28.39	28064.26
	5	27.34	12.94	26.53	28.14	27254.84
	6	28.22	12.76	27.42	29.02	27766.52
	7	27.12	13.04	26.32	27.91	28038.74
	8	27.50	13.10	26.69	28.30	27965.62
	9	27.62	12.79	26.83	28.41	27949.15
	10	27.18	12.95	26.37	27.99	26609.25

ANOVA for Smoking Prevalence by Family Background

The ANOVA Procedure

Dependent Variable: Smoking_Prevalence

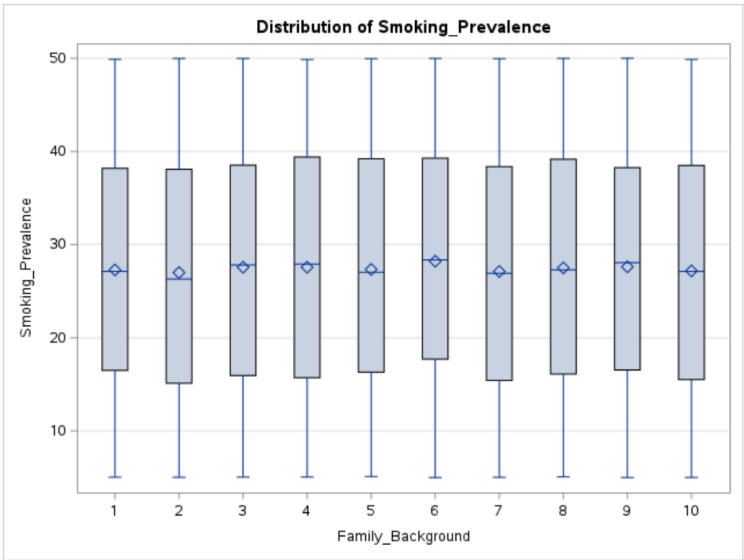
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	1060.641	117.849	0.70	0.7097
Error	9990	1682414.147	168.410		
Corrected Total	9999	1683474.788			

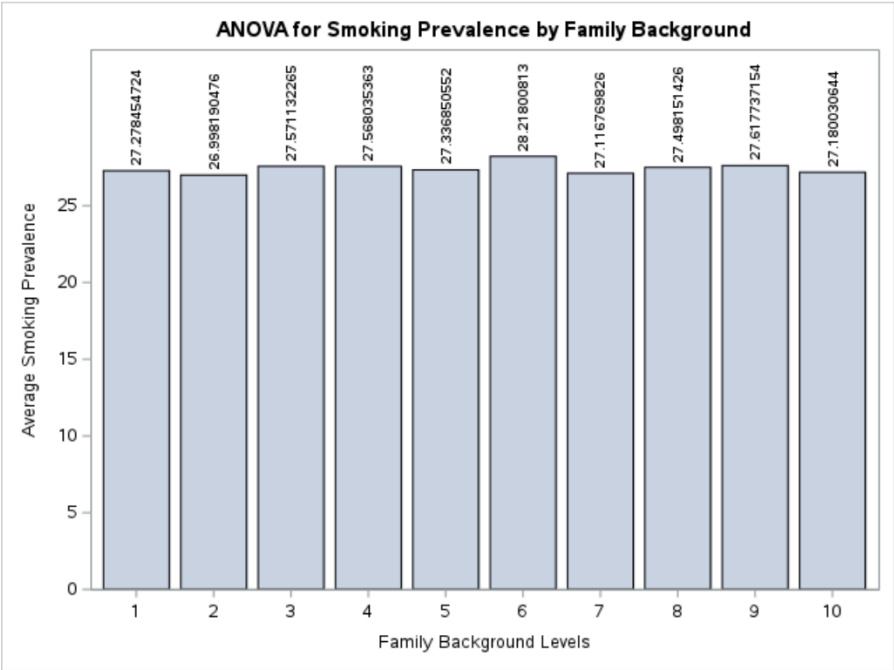
R-Square	Coeff Var	Root MSE	Smoking_Prevalence Mean
0.000630	47.29458	12.97728	27.43926

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Family_Background	9	1060.641449	117.849050	0.70	0.7097

ANOVA for Smoking Prevalence by Family Background

The ANOVA Procedure





Smoking Prevalence by Parental Supervision

Summary Statistics for Parental Supervision		Mean	Std	95_LCLM	95_UCLM	All
Smoking_Prevalence	Parental_Supervision					Sum
	1	27.53	12.84	26.74	28.31	28627.46
	2	26.76	12.73	25.96	27.57	25774.29
	3	27.44	13.19	26.61	28.27	26613.83
	4	27.36	13.11	26.54	28.19	26488.71
	5	27.68	12.71	26.90	28.46	28148.21
	6	27.87	13.09	27.06	28.68	28146.76
	7	27.85	13.22	27.02	28.67	27317.14
	8	27.26	13.02	26.44	28.08	26358.14
	9	27.00	12.88	26.22	27.78	28509.31
	10	27.63	12.99	26.84	28.43	28408.72

ANOVA for Smoking Prevalence by Parental Supervision

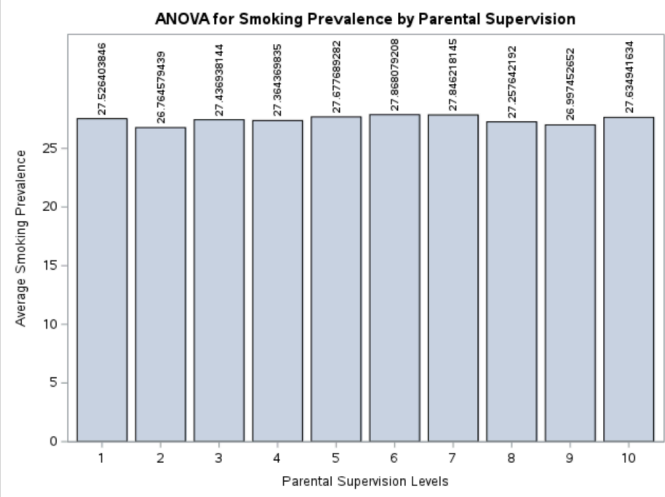
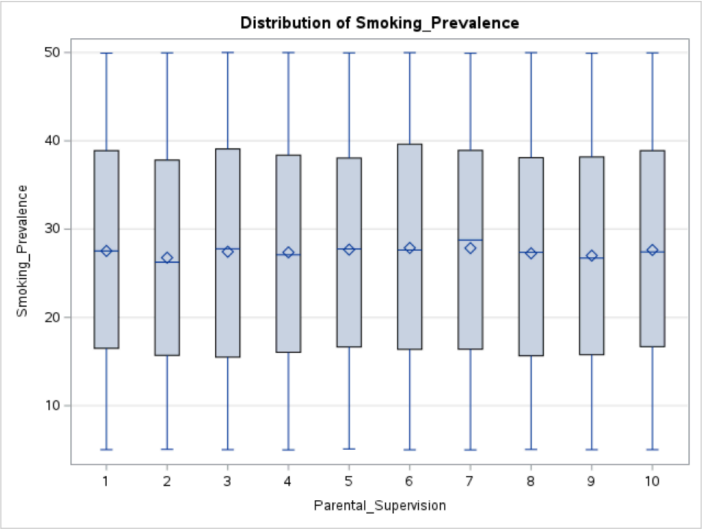
The ANOVA Procedure

Dependent Variable: Smoking_Prevalence

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	9	1135.076	126.120	0.75	0.6641
Error	9990	1682339.712	168.402		
Corrected Total	9999	1683474.788			

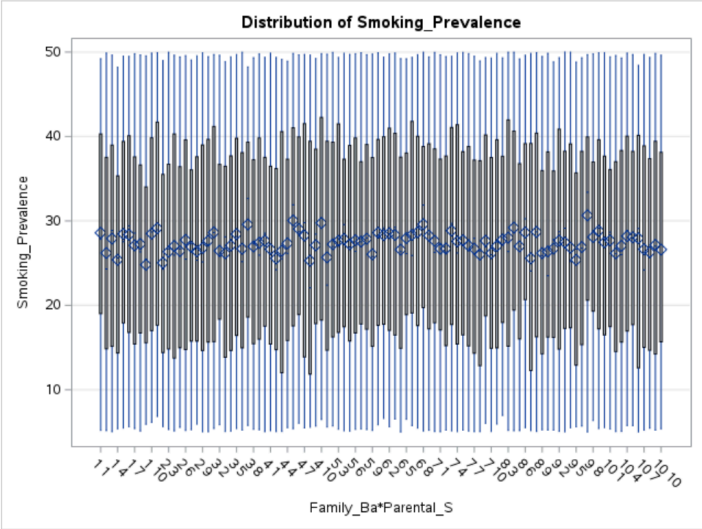
R-Square	Coeff Var	Root MSE	Smoking_Prevalence Mean
0.000674	47.29353	12.97699	27.43926

Source	DF	Anova SS	Mean Square	F Value	Pr > F
Parental_Supervision	9	1135.076321	126.119591	0.75	0.6641



Interaction of Family Background and Parental Supervision on Smoking Prevalence

The GLM Procedure



Interaction of Family Background and Parental Supervision on Smoking Prevalence

The GLM Procedure

Class Level Information		
Class	Levels	Values
Family_Background	10	1 2 3 4 5 6 7 8 9 10
Parental_Supervision	10	1 2 3 4 5 6 7 8 9 10

Number of Observations Read	10000
Number of Observations Used	10000

Interaction of Family Background and Parental Supervision on Smoking Prevalence

The GLM Procedure

Dependent Variable: Smoking_Prevalence

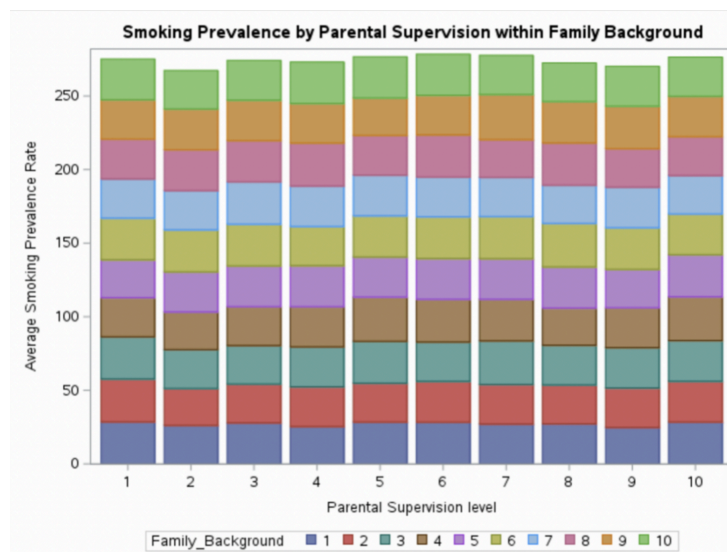
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	99	13150.215	132.830	0.79	0.9411
Error	9900	1670324.574	168.720		
Corrected Total	9999	1683474.788			

R-Square	Coeff Var	Root MSE	Smoking_Prevalence Mean
----------	-----------	----------	-------------------------

R-Square	Coeff Var	Root MSE	Smoking_Prevalence Mean
0.007811	47.33806	12.98921	27.43926

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Family_Background	9	1060.64145	117.84905	0.70	0.7109
Parental_Supervision	9	1131.73170	125.74797	0.75	0.6675
Family_Ba*Parental_S	81	10957.84156	135.28199	0.80	0.9029

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Family_Background	9	1085.85081	120.65009	0.72	0.6956
Parental_Supervision	9	1073.91927	119.32436	0.71	0.7029
Family_Ba*Parental_S	81	10957.84156	135.28199	0.80	0.9029



Graph 3-5: The Smoking Prevalence by Parental Supervision within Family Background

d. Specific Question 4

The relationship between mental health and drug experimentation among young people is complicated. We all know that mental health issues such as depression, anxiety will cause some bad coping methods such as alcohol and furthermore, drugs. Other than that, socioeconomic status further influences this dynamic. The ability to reach more resources can affect a lot. Hence, we want to focus on the question: How does mental health status influence drug experimentation rates among youth with different socioeconomic levels?

Methods:

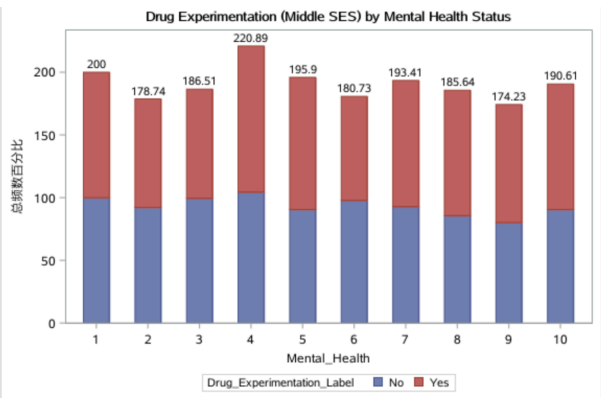
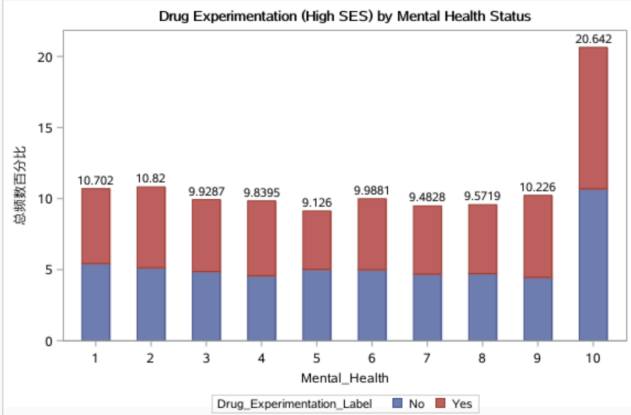
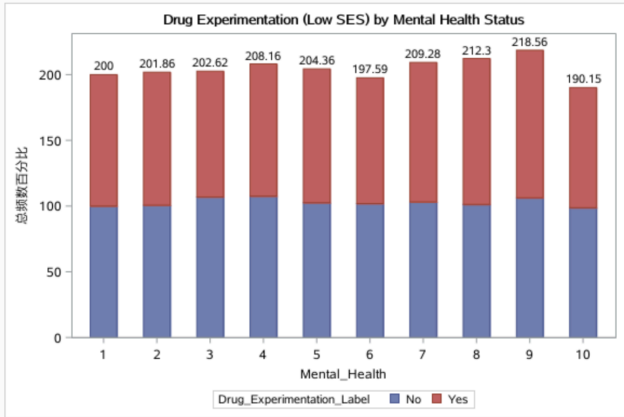
First, I conducted EDA to explore some potential relationships between mental health and drug experimentation (with the threshold of mean value 40 and above is considered “drug-use”), but it turned out that the relationship was not clear so further analysis was conducted. I proceeded with a Chi-Square Test of Independence to evaluate whether a significant association exists between these two variables with the null hypothesis : there is no significant relationship between the drug experimentation and the mental health problems.

Results:

The p-value of high and middle SES for those two variables is 0.28, and the low SES is 0.96. All 3 of the values are greater than 0.05, meaning we fail to reject the null hypothesis, so there is not a significant relationship between those variables no matter how the SES changes.

The results showed that there is no significant relationship between mental health and the drug experiment, meaning we need to dig more into other factors that may affect the drug experimentation.

Here are the pictures of EDA:



Drug Experimentation (Middle SES) by Mental Health Status

FREQ procedure

The y-axis is the percentage. We see that the distribution is pretty even.
Here are the pictures for Chisq Analysis:

Chi-Square Test for Drug Experimentation and Mental Health (High SES)

FREQ procedure

Frequency
percentage
Row Percent
Column Percent

Drug_Experimentation_Label-Mental_Health table											
Drug_Experimentation_Label	Mental_Health										total
	1	2	3	4	5	6	7	8	9	10	
No	183	173	164	154	169	168	158	159	150	177	1655
	5.44	5.14	4.88	4.58	5.02	4.99	4.70	4.73	4.46	5.26	49.20
	11.06	10.45	9.91	9.31	10.21	10.15	9.55	9.61	9.06	10.69	
	50.83	47.53	49.10	46.53	55.05	50.00	49.53	49.38	43.60	51.01	
Yes	177	191	170	177	138	168	161	163	194	170	1709
	5.26	5.68	5.05	5.26	4.10	4.99	4.79	4.85	5.77	5.05	50.80
	10.36	11.18	9.95	10.36	8.07	9.83	9.42	9.54	11.35	9.95	
	49.17	52.47	50.90	53.47	44.95	50.00	50.47	50.62	56.40	48.99	
total	360	364	334	331	307	336	319	322	344	347	3364
	10.70	10.82	9.93	9.84	9.13	9.99	9.48	9.57	10.23	10.32	100.00

Statistics for the table "Mental_Health-Drug_Experimentation_Label"

Statistics	Degrees of Freedom	value	Probability
Chi-square	9	10.8094	0.2890
Likelihood ratio chi-square test	9	10.8290	0.2876
Mantel-Haenszel Chi-Square	1	0.1688	0.6812
Phi coefficient		0.0567	
Series Connection Coefficient		0.0566	
Cramer V		0.0567	

Sample size = 3364

Chi-Square Test for Drug Experimentation and Mental Health (Middle SES)

FREQ procedure

Drug_Experimentation_Label-Mental_Health table											
Drug_Experimentation_Label	Mental_Health										total
	1	2	3	4	5	6	7	8	9	10	
No	181	167	180	189	164	177	168	155	145	164	1690
	5.41	4.99	5.38	5.65	4.90	5.29	5.02	4.64	4.34	4.90	50.54
	10.71	9.88	10.65	11.18	9.70	10.47	9.94	9.17	8.58	9.70	
Yes	51.57	53.18	54.88	48.84	47.81	55.66	49.56	47.69	47.54	49.10	
	170	147	148	188	179	141	171	160	170	1654	
	5.08	4.40	4.43	5.92	5.35	4.22	5.11	5.08	4.78	5.08	49.46
total	10.28	8.89	8.95	11.97	10.82	8.52	10.34	10.28	9.67	10.28	
	48.43	46.82	45.12	51.16	52.19	44.34	50.44	52.31	52.46	50.90	
	351	314	328	387	343	318	339	325	305	334	3344
	10.50	9.39	9.81	11.57	10.26	9.51	10.14	9.72	9.12	9.99	100.00

Statistics for the table "Mental_Health-Drug_Experimentation_Label"

Statistics	Degrees of Freedom	value	Probability
Chi-square	9	10.8594	0.2855
Likelihood ratio chi-square test	9	10.8735	0.2845
Mantel-Haenszel Chi-Square	1	2.8658	0.0905
Phi coefficient		0.0570	
Series Connection Coefficient		0.0569	
Cramer V		0.0570	

Sample size = 3344

Chi-Square Test for Drug Experimentation and Mental Health (Low SES)

FREQ procedure

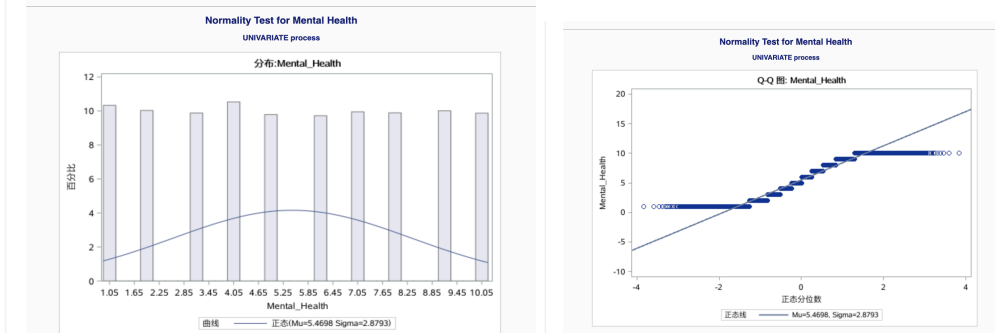
Drug_Experimentation_Label-Mental_Health table											
Drug_Experimentation_Label	Mental_Health										total
	1	2	3	4	5	6	7	8	9	10	
No	159	160	170	171	163	162	164	161	169	157	1636
	4.83	4.86	5.16	5.19	4.95	4.92	4.98	4.89	5.13	4.77	49.70
	9.72	9.78	10.39	10.45	9.96	9.90	10.02	9.84	10.33	9.60	
Yes	49.38	49.23	52.15	51.04	49.54	50.94	48.66	47.08	48.01	51.31	
	163	165	166	164	166	156	173	181	183	149	1656
	4.95	5.01	4.74	4.98	5.04	4.74	5.26	5.50	5.56	4.53	50.30
total	9.84	9.96	9.42	9.90	10.02	9.42	10.45	10.93	11.05	9.00	
	50.62	50.77	47.85	48.96	50.46	49.06	51.34	52.92	51.99	48.69	
	322	325	326	335	329	318	337	342	352	306	3292
	9.78	9.87	9.90	10.18	9.99	9.66	10.24	10.39	10.69	9.30	100.00

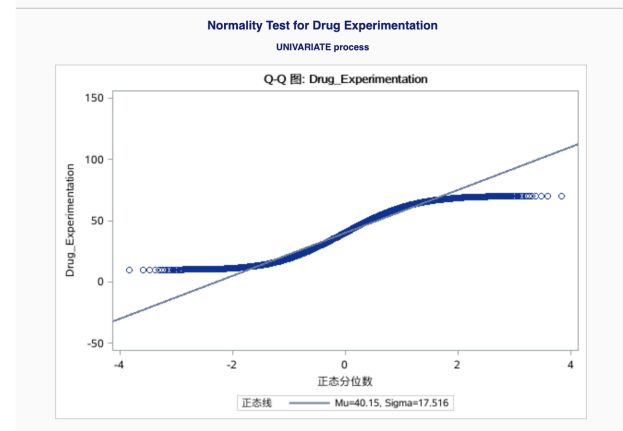
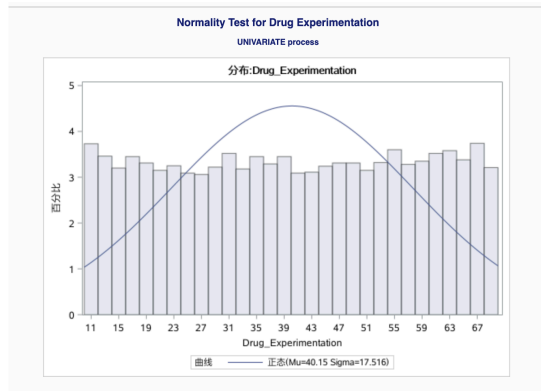
Statistics for the table "Mental_Health-Drug_Experimentation_Label"

Statistics	Degrees of Freedom	value	Probability
Chi-square	9	3.0692	0.9615
Likelihood ratio chi-square test	9	3.0702	0.9614
Mantel-Haenszel Chi-Square	1	0.2481	0.6184
Phi coefficient		0.0305	
Series Connection Coefficient		0.0305	
Cramer V		0.0305	

Sample size = 3292

I also checked the assumptions and everything looks great:





4. Discussions

The results showed that the drug experimentation among young people are significantly influenced by parental supervision and family stability emerging. Also, peer networks and environmental stressors may also play a role and require further investigation. However, there do exist some limitations of the analysis. For example, the simple categories of socioeconomic status and mental health status may affect the analysis and large sample size might not be generalizable among diverse populations. Hence, in the future, we can investigate more factors that may influence the mental health problems such as peer pressure, school environment in order to get the youth drug consumption reduced.

5. References

Ali, Waqar. "Youth Smoking and Drug Dataset 🚭💊." Kaggle, October 8, 2024.
<https://www.kaggle.com/datasets/waqi786/youth-smoking-and-drug-dataset>.

SAS Institute, Inc. (2018) SAS Studio 3.8 (Enterprise Edition) for Linux [Computer software]. Cary, NC.